Right-Triangle Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Trigonometric ratios**

A right triangle has legs of length 5 and 12 and hypotenuse 13. For the angle θ opposite the side of length 5, find:

a $\sin\theta = \frac{5}{13}$ **b** $\cos\theta = \frac{12}{13}$ **c** $\tan\theta = \frac{5}{12}$

(a) $= \frac{5}{13}$; (b) $= \frac{12}{13}$; (c) $= \frac{5}{12}$

2 **[Foundational]** • **Target: Trigonometric ratios**

Write the exact value of each expression:

a $\sin 30° = \frac{1}{2}$ **b** $\cos 45° = \frac{\sqrt{2}}{2}$ **c** $\tan 60° = \sqrt{3}$ **d** $\sin 60° = \frac{\sqrt{3}}{2}$

(a) $= \frac{1}{2}$; (b) $= \frac{\sqrt{2}}{2}$; (c) $= \sqrt{3}$; (d) $= \frac{\sqrt{3}}{2}$

3 **[Foundational]** • **Target: Trigonometric ratios**

A right triangle has legs of length 7 and 24.

a Find the length of the hypotenuse. $\sqrt{7^2 + 24^2} = \sqrt{625} = 25$.

b Find the measure of the angle opposite the side of length 7, to the nearest hundredth of a degree. $\theta = \tan^{-1}\left(\frac{7}{24}\right) \approx 16.26°$.

(a) $\sqrt{7^2 + 24^2} = \sqrt{625} = 25$.; (b) $\theta = \tan^{-1}\left(\frac{7}{24}\right) \approx 16.26°$.

4 **[Foundational]** • **Target: Applications**

A ladder leans against a wall, making an angle of $62°$ with the ground. The foot of the ladder is 4.5 m from the base of the wall. How high up the wall does the ladder reach? Round to two decimal places.

$h = 4.5\tan 62° \approx 4.5 \times 1.8807 \approx 8.46$ m.

5 **[Foundational]** • **Target: Applications**

From the top of a cliff 50 m above sea level, the angle of depression to a boat is $18°$. How far is the boat from the base of the cliff? Round to one decimal place.

$d = \frac{50}{\tan 18°} \approx \frac{50}{0.3249} \approx 153.9$ m.

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6 [Foundational] • Target: Applications

A kite string is 85 m long and makes an angle of $54°$ with the horizontal ground. Assuming the string is straight, find the height of the kite above the ground, to the nearest metre.

$h = 85\sin 54° \approx 85 \times 0.8090 \approx 69$ m.

7 [Foundational] • Target: Applications

Two buildings are 40 m apart. From the top of the shorter building (height 25 m), the angle of elevation to the top of the taller building is $32°$. Find the height of the taller building, to one decimal place.

Rise $= 40\tan 32° \approx 40(0.6249) \approx 25.0$ m, so the taller building is about $25 + 25.0 = 50.0$ m.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Applications

A surveyor stands 120 m from the base of a tower and measures the angle of elevation to the top as $37°$. Find the height of the tower, to the nearest tenth of a metre.

$h = 120\tan 37° \approx 120(0.7536) \approx 90.4$ m.

9 [Intermediate] • Target: Solving right triangles

In right triangle ABC with the right angle at C, $A = 35°$ and the hypotenuse $AB = 20$. Find:

a side BC (opposite A) $BC = 20\sin 35° \approx 11.47$.

b side AC (adjacent to A) $AC = 20\cos 35° \approx 16.38$.

c angle B $B = 90° - 35° = 55°$.

(a) $BC = 20\sin 35° \approx 11.47$.; (b) $AC = 20\cos 35° \approx 16.38$.; (c) $B = 90° - 35° = 55°$.

10 [Intermediate] • Target: Solving right triangles

In right triangle PQR with the right angle at R, the two legs are $PR = 9$ and $QR = 12$. Find angle P (opposite QR) to the nearest tenth of a degree, and the length of the hypotenuse.

$\tan P = \frac{12}{9}$, so $P = \tan^{-1}\left(\frac{12}{9}\right) \approx 53.1°$. Hypotenuse $= \sqrt{9^2 + 12^2} = 15$.

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11 [Intermediate] • Target: Angles of elevation and depression

A drone hovers at a height of 60 m. The angle of depression from the drone to a landmark on the ground is $27°$. Find the horizontal distance from the point directly below the drone to the landmark, to the nearest metre.

$d = \frac{60}{\tan 27°} \approx \frac{60}{0.5095} \approx 118$ m.

12 [Intermediate] • Target: Angles of elevation and depression

From a lighthouse 45 m tall, the angle of depression to a ship is $12°$, and a second ship is spotted further out with angle of depression $7°$. Find the distance between the two ships, to the nearest metre.

Distances from the base: $d_1 = \frac{45}{\tan 12°} \approx 211.7$ m; $d_2 = \frac{45}{\tan 7°} \approx 366.5$ m. Distance between ships $\approx 366.5 - 211.7 \approx 154.8$ m.

13 [Intermediate] • Target: The reciprocal trig ratios

For a right triangle with opposite $= 3$, adjacent $= 4$, hypotenuse $= 5$ (relative to angle θ), find:

a $\csc\theta = \frac{5}{3}$ **b** $\sec\theta = \frac{5}{4}$ **c** $\cot\theta = \frac{4}{3}$

(a) $= \frac{5}{3}$; (b) $= \frac{5}{4}$; (c) $= \frac{4}{3}$

14 [Intermediate] • Target: The reciprocal trig ratios

Explain why $\csc\theta = \frac{1}{\sin\theta}$, and use this to find $\csc(30°)$.

$\csc\theta$ is defined as the reciprocal of $\sin\theta$ (hypotenuse over opposite, instead of opposite over hypotenuse). $\csc(30°) = \frac{1}{\sin(30°)} = \frac{1}{1/2} = 2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Multi-step surveying problems • Cross-Lesson: 6.02 Polar Equations and Graphs

A surveyor at point A measures the angle of elevation to a mountain peak as $22°$. Walking 500 m directly toward the mountain to point B, the angle of elevation increases to $35°$. Find the height of the mountain, to the nearest metre. (Hint: let h be the height and x the distance from B to the base; set up two equations.)

From B: $h = x\tan 35°$. From A: $h = (x + 500)\tan 22°$. Setting equal: $x\tan 35° = (x + 500)\tan 22°$. $x(0.7002) = (x + 500)(0.4040)$. $0.7002x - 0.4040x = 202$. $0.2962x = 202 \Rightarrow x \approx 682.0$. $h = 682.0(0.7002) \approx 477.5$ m.

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16 **[Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 7.01 Exponential Functions and Their Graphs**

In a 30-60-90 triangle, the shortest side (opposite the $30°$ angle) has length 5. Find the lengths of the other two sides in exact form.

Side ratios are $1 : \sqrt{3} : 2$. Side opposite $60°$: $5\sqrt{3}$. Hypotenuse: 10.

17 **[Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 7.02 Logarithms and Their Properties**

In a 45-45-90 triangle, each leg has length 7. Find the hypotenuse in exact form.

Hypotenuse $= 7\sqrt{2}$ (using the ratio $1 : 1 : \sqrt{2}$).

18 **[Challenging] • Target: Trigonometric ratios beyond the first quadrant (preview) • Cross-Lesson: 7.03 Exponential and Logarithmic Equations**

A right triangle used to define $\sin\theta$ and $\cos\theta$ only works directly for acute angles θ. Explain why a different (coordinate-based) definition is needed to extend these ratios to obtuse or negative angles.

A right triangle can only have angles strictly between $0°$ and $90°$ (since it already has one $90°$ angle and angles sum to $180°$), so triangle ratios cannot directly define sine or cosine for angles at or beyond $90°$. Extending to all angles requires the unit-circle definition, where $\cos\theta$ and $\sin\theta$ are the x- and y-coordinates of a point on the unit circle at angle θ.

19 **[Challenging] • Target: Solving right triangles • Cross-Lesson: 7.04 Applications: Growth and Decay**

In ΔABC, $a = 8$, $b = 11$, and $A = 35^\circ$. Determine the number of possible triangles and find the measure of angle B for all valid cases to the nearest tenth of a degree.

Calculate the altitude $h = b\sin A = 11\sin 35^\circ \approx 11(0.5736) = 6.31$. Since $htwopossibletriangles. BytheLawofSines$: \sin B = \frac{b\sin A}{a} = \frac{6.31}{8} \approx 0.7887.x. Second solution (obtuse): $B_2 = 180^\circ - 52.1^\circ = 127.9^\circ$. Check validity: $A + B_2 = 35^\circ + 127.9^\circ = 162.9^\circ 0$. Both $B_1 \approx 52.1^\circ$ and $B_2 \approx 127.9^\circ$ are valid.

20 **[Challenging] • Target: Angles of elevation and depression • Cross-Lesson: 8.01 Rational Functions and Asymptotes**

A surveyor measures distances from an observation point C to two landmarks A and B: $AC = 140$ m and $BC = 210$ m, with included angle $\angle ACB = 78^\circ$. Find the distance AB between the landmarks to the nearest tenth of a meter, and compute the area of ΔABC to the nearest square meter.

Apply the Law of Cosines to find side $c = AB$: $c^2 = a^2 + b^2 - 2ab\cos C = 210^2 + 140^2 - 2(210)(140)\cos 78^\circ$. $c^2 = 44100 + 19600 - 58800(0.2079) = 63700 - 12225 = 51475$. $c = \sqrt{51475} \approx 226.9$ m. Compute the area: Area $= \frac{1}{2}ab\sin C = \frac{1}{2}(210)(140)\sin 78^\circ = 14700(0.9781) \approx 14379$ m^2.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Trigonometric ratios

In a right-angled triangle ABC, where the right angle is at C, if $AC = 10$ and $BC = 24$, find the length of the hypotenuse AB.

a. 25

b. **26** ✓

c. 27

d. 28

We can use the Pythagorean theorem to find the length of the hypotenuse in a right-angled triangle. The theorem states that $a^2 + b^2 = c^2$, where a and b are the lengths of the two shorter sides (legs) and c is the length of the hypotenuse. In this triangle, $AC = 10$ and $BC = 24$. Therefore, we have: $AB^2 = AC^2 + BC^2$ $AB^2 = 10^2 + 24^2$ $AB^2 = 100 + 576$ $AB^2 = 676$ $AB = \sqrt{676}$ $AB = 26$ The length of the hypotenuse AB is 26.

22 [Foundational] • Target: Applications

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$ ✓

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

We are given $\cos\alpha = \frac{3}{5}$ and α is in the first quadrant. Using the identity $\sin^2\alpha + \cos^2\alpha = 1$, we find $\sin\alpha = \sqrt{1 - (\frac{3}{5})^2} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$ (since α is in the first quadrant, $\sin\alpha$ is positive). We are given $\sin\beta = -\frac{4}{5}$ and β is in the third quadrant. Using the identity $\sin^2\beta + \cos^2\beta = 1$, we find.

23 [Foundational] • Target: Solving right triangles

In triangle ABC, the lengths of sides a, b, and c are opposite to angles A, B, and C respectively. Given that $a = 7$, $b = 10$, and angle $C = 60^\circ$, find the exact length of side c.

a. $\sqrt{79}$

b. 9

c. $\sqrt{91}$ ✓

d. 10

We can use the Law of Cosines to find the length of side c. The Law of Cosines states that $c^2 = a^2 + b^2 - 2ab\cos(C)$. Substituting the given values: $c^2 = 7^2 + 10^2 - 2(7)(10)\cos(60^\circ)$ $c^2 = 49 + 100 - 140(\frac{1}{2})$ $c^2 = 149 - 70$ $c^2 = 79$ Therefore, $c = \sqrt{79}$.

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24 [Foundational] • Target: Angles of elevation and depression

If $\cos\theta = \frac{1}{2}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1}{4}$

b. $\frac{1}{2}$ ✓

c. $\frac{3}{5}$

d. $-\frac{1}{2}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{1}{2}\right)}{2}} = \frac{1}{2}$.

25 [Foundational] • Target: The reciprocal trig ratios

A complex number z is represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Given that $z^3 = 8i$, and z lies in the first quadrant such that its argument θ satisfies $0 < \theta < \frac{\pi}{2}$. Find the value of $r^2\cos\theta$.

a. 2

b. **4** ✓

c. $2\sqrt{2}$

d. 1

We are given $z^3 = 8i$. In polar form, $8i = 8(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$. Let $z = r(\cos\theta + i\sin\theta)$. Then $z^3 = r^3(\cos(3\theta) + i\sin(3\theta))$. Equating the magnitudes: $r^3 = 8$, so $r = 2$. Equating the arguments: $3\theta = \frac{\pi}{2} + 2k\pi$ for integer k. Since z is in the first quadrant, $0 < \theta < \frac{\pi}{2}$. For $k = 0$, $3\theta = \frac{\pi}{2}$, so $\theta = \frac{\pi}{6}$. This lies in the first quadrant. For $k = 1$.

26 [Foundational] • Target: Multi-step surveying problems

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

The problem states that point $P(x, y)$ lies on the unit circle, meaning $x^2 + y^2 = 1$. We are given that the y-coordinate of P is $y = \frac{3}{5}$. Substituting this into the unit circle equation, we get $x^2 + (\frac{3}{5})^2 = 1$. This simplifies to $x^2 + \frac{9}{25} = 1$. Subtracting $\frac{9}{25}$ from both sides gives $x^2 = 1 - \frac{9}{25} = \frac{16}{25}$. Taking the square root of both sides, we get $x = \pm\frac{4}{5}$. The problem also states that point P is in the second quadrant, where the x.

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27 [Foundational] • Target: The 30-60-90 and 45-45-90 special triangles

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$ ✓

d. $\frac{4}{5}$

On the unit circle, any point P(x, y) satisfies the equation $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $\frac{3}{5}$ and P is in the second quadrant. For a point $P(\cos\theta, \sin\theta)$ on the unit circle, we have $x = \cos\theta$ and $y = \sin\theta$. Thus, we have $\sin\theta = \frac{3}{5}$. Substituting this into the unit circle equation: $\cos^2\theta + (\frac{3}{5})^2 = 1$. $\cos^2\theta + \frac{9}{25} = 1$. $\cos^2\theta = 1 - \frac{9}{25}$. $\cos^2\theta = \frac{16}{25}$..

28 [Foundational] • Target: Trigonometric ratios beyond the first quadrant (preview)

If $\sin\theta = \frac{5}{13}$ and θ is in Quadrant II, find $\cos\theta$.

a. $\frac{5}{13}$

b. $-\frac{12}{13}$ ✓

c. $-\frac{107}{130}$

d. $\frac{12}{13}$

Using the Pythagorean triple $5-12-13$ with Quadrant II signs: $\cos\theta = -\frac{12}{13}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Trigonometric ratios

If $\cos\theta = -\frac{15}{17}$ and θ is in Quadrant III, find $\sin\theta$.

a. $-\frac{63}{170}$

b. $\frac{8}{17}$

c. $\frac{15}{17}$

d. $-\frac{8}{17}$ ✓

Using the Pythagorean triple $8-15-17$ with Quadrant III signs: $\sin\theta = -\frac{8}{17}$.

30 [Intermediate] • Target: Applications

If $\cos\theta = \frac{24}{25}$ and θ is in Quadrant IV, find $\sin\theta$.

a. $-\frac{7}{25}$ ✓

b. $\frac{24}{25}$

c. $-\frac{9}{50}$

d. $\frac{7}{25}$

Using the Pythagorean triple $7-24-25$ with Quadrant IV signs: $\sin\theta = -\frac{7}{25}$.

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31 [Intermediate] • Target: Solving right triangles

If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant I, find $\cos\theta$.

a. $\frac{4}{5}$ ✓

b. $\frac{3}{5}$

c. $\frac{9}{10}$

d. $-\frac{4}{5}$

Using the Pythagorean triple $3-4-5$ with Quadrant I signs: $\cos\theta = \frac{4}{5}$.

32 [Intermediate] • Target: Angles of elevation and depression

If $\cos\theta = \frac{4}{5}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{4}{5}$

b. $-\frac{3}{5}$

c. $\frac{3}{5}$ ✓

d. $\frac{7}{10}$

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{16}{25}} = \frac{3}{5}$.

33 [Intermediate] • Target: The reciprocal trig ratios

In a right-angled triangle, one of the acute angles measures 30°. If the side opposite this angle has a length of 5 units, what is the length of the hypotenuse?

a. **10 units** ✓

b. 5 units

c. $5\sqrt{3}$ units

d. 7.5 units

Let the right-angled triangle be denoted as ABC, with the right angle at C. Let angle A be 30°. The side opposite to angle A is BC, and its length is given as 5 units. We are looking for the length of the hypotenuse AB. The sine of an angle in a right-angled triangle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. Therefore, $\sin(A) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{BC}{AB}$. We have $\sin(30^\circ) = \frac{5}{AB}$. We know that.

34 [Intermediate] • Target: Multi-step surveying problems

In a right-angled triangle, one of the acute angles is 30°. What is the measure of the other acute angle?

a. 30°

b. 45°

c. 60° ✓

d. 90°

In a right-angled triangle, the sum of the three angles is 180°. One angle is 90° (the right angle), and another acute angle is given as 30°. Let the other acute angle be x. Therefore, we have the equation: $90^\circ + 30^\circ + x = 180^\circ$. Subtracting 120° from both sides gives $x = 180^\circ - 120^\circ = 60^\circ$.

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35 **[Intermediate]** • **Target: The 30-60-90 and 45-45-90 special triangles**

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?

a. $\cos\theta$ ✓

b. $\sin\theta$

c. $\tan\theta$

d. 1

By definition, the coordinates of a point $P(x, y)$ on the unit circle corresponding to an angle θ (measured counterclockwise from the positive x-axis) are given by $(\cos\theta, \sin\theta)$. Therefore, the x-coordinate of the point P is $\cos\theta$.

36 **[Intermediate]** • **Target: Trigonometric ratios beyond the first quadrant (preview)**

In ΔABC, $\angle A = 30^\circ$, $\angle B = 90^\circ$, and side $a = 10$. Find side b (Law of Sines).

a. 10

b. 22

c. 5

d. **20** ✓

$\frac{a}{\sin A} = \frac{b}{\sin B} \Rightarrow b = \frac{10\sin 90^\circ}{\sin 30^\circ} = 20.$

37 **[Intermediate]** • **Target: Trigonometric ratios**

Let the complex number z be represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Suppose z satisfies the equation $z^3 = -64i$. If z is restricted to the fourth quadrant (i.e., $\frac{3\pi}{2} < \arg(z) < 2\pi$), find the real part of z.

a. 4

b. -4

c. $2\sqrt{3}$ ✓

d. $-2\sqrt{3}$

We are given the equation $z^3 = -64i$. First, express $-64i$ in polar form. The modulus is $|-64i| = 64$. The argument is $\arg(-64i) = \frac{3\pi}{2}$ (or $-\frac{\pi}{2}$). So, $-64i = 64\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right)$. Let $z = r(\cos\phi + i\sin\phi)$. Then $z^3 = r^3(\cos(3\phi) + i\sin(3\phi))$. Equating the polar forms, we have $r^3 = 64$ and $3\phi = \frac{3\pi}{2} + 2k\pi$ for some integer k. From $r^3 = 64$, we get $r = 4$ (since r is a non-negative real number). Fro.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Applications • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Consider a circle with the equation $x^2 + y^2 = 25$. A point $P(x_0, y_0)$ lies on this circle. A tangent line to the circle at point P intersects the x-axis at point A and the y-axis at point B. If the area of the triangle OAB, where O is the origin $(0, 0)$, is minimized, find the value of $x_0^2 + y_0^2$.

a. **25** ✓

b. 50

c. 100

d. 125

The equation of the circle is $x^2 + y^2 = 25$. Let the point of tangency be $P(x_0, y_0)$. The equation of the tangent line to the circle at $P(x_0, y_0)$ is given by $xx_0 + yy_0 = 25$. To find the x-intercept A, we set $y = 0$: $xx_0 = 25 \Rightarrow x = \frac{25}{x_0}$. So, $A = (\frac{25}{x_0}, 0)$. To find the y-intercept B, we set $x = 0$: $yy_0 = 25 \Rightarrow y = \frac{25}{y_0}$. So, $B = (0, \frac{25}{y_0})$. The area of triangle OAB is.

39 [Challenging] • Target: Solving right triangles • Cross-Lesson: 7.04 Applications: Growth and Decay

Consider the complex number $z = \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{10}$. If z is expressed in polar form as $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $0 \le \theta < 2\pi$, find the exact values of r and θ.

a. $r = 1, \theta = \frac{5\pi}{3}$ ✓

b. $r = 1, \theta = \frac{4\pi}{3}$

c. $r = 10, \theta = \frac{5\pi}{3}$

d. $r = \sqrt{3}, \theta = \frac{4\pi}{3}$

First, express the base complex number in polar form. Let $w = \frac{\sqrt{3}}{2} + \frac{1}{2}i$. The modulus of w is $|w| = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{3}{4} + \frac{1}{4}} = \sqrt{1} = 1$. The argument of w is $\arg(w) = \arctan\left(\frac{1/2}{\sqrt{3}/2}\right) = \arctan\left(\frac{1}{\sqrt{3}}\right)$. Since the real and imaginary parts are both positive, w is in the first quadrant, so $\arg(w) = \frac{\pi}{6}$. Thus.

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40 [Challenging] • Target: Angles of elevation and depression • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n?

a. $a_n = 5 + 3(n - 1)$ ✓

b. $a_n = 3n + 2$

c. $a_n = 5n + 3$

d. $a_n = 5 + 3n$

The formula for the n-th term of an arithmetic sequence is given by $a_n = a_1 + d(n - 1)$, where a_1 is the first term and d is the common difference. In this case, $a_1 = 5$ and $d = 3$. Substituting these values into the formula gives $a_n = 5 + 3(n - 1)$. If we simplify this, we get $a_n = 5 + 3n - 3 = 3n + 2$. Therefore, the direct formula is $a_n = 5 + 3(n - 1)$ or its simplified form $a_n = 3n + 2$. Choice A directly uses the definition. Choice B is the simplified form. Let's examine all options: Choice A.

41 [Challenging] • Target: The reciprocal trig ratios • Cross-Lesson: 8.02 Graphing Rational Functions

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

a. 4

b. **5** ✓

c. 6

d. 7

The sum of the first n terms of a geometric sequence is given by the formula $S_n = a_1\frac{r^n - 1}{r - 1}$, where a_1 is the first term and r is the common ratio. Given $a_1 = 5$ and $r = 2$, and $S_n = 315$, we can substitute these values into the formula: $315 = 5\frac{2^n - 1}{2 - 1}$ $315 = 5\frac{2^n - 1}{1}$ $315 = 5(2^n - 1)$ Divide both sides by 5: $\frac{315}{5} = 2^n - 1$ $63 = 2^n - 1$ Add 1 to both sides: $63 + 1 = 2^n$ $64 = 2^n$ Since $2^6 = 64$, we have $n = 6$. Let's check the choices.

42 [Challenging] • Target: Multi-step surveying problems • Cross-Lesson: 8.03 Introduction to Limits

In a right-angled triangle ABC, angle B is 90°. If $AB = 5$ and $BC = 12$, what is the length of the hypotenuse AC?

a. 10

b. 11

c. **13** ✓

d. 17

This problem involves finding the hypotenuse of a right-angled triangle given the lengths of the other two sides. We can use the Pythagorean theorem, which states that in a right-angled triangle, the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides. Let a and b be the lengths of the two shorter sides, and c be the length of the hypotenuse. The Pythagorean theorem is given by: $a^2 + b^2 = c^2$. In triangle ABC, we have: $AB = a = 5$.

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43 [Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 9.01 Arithmetic Sequences and Series

In a right-angled triangle ABC, where angle B is 90 degrees, if AB = 3 and BC = 4, what is the length of the hypotenuse AC?

a. 7

b. 5 ✓

c. 12

d. 25

This question involves a right-angled triangle and requires the application of the Pythagorean theorem. The Pythagorean theorem states that in a right-angled triangle, the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides. In triangle ABC, AC is the hypotenuse. According to the theorem: $AC^2 = AB^2 + BC^2$ Substituting the given values: $AC^2 = 3^2 + 4^2$ $AC^2 = 9 + 16$ $AC^2 = 25$ To find the length of AC, we take the square root of both sides.

44 [Challenging] • Target: Trigonometric ratios beyond the first quadrant (preview) • Cross-Lesson: 9.02 Geometric Sequences and Series

A right-angled triangle ABC is shown with the right angle at vertex C. The length of side AC is 8 units and the length of side BC is 6 units. What is the value of $\sin(\angle BAC)$?

a. $\frac{3}{5}$ ✓

b. $\frac{4}{5}$

c. $\frac{3}{4}$

d. $\frac{4}{3}$

The triangle ABC is a right-angled triangle with the right angle at C. We are given AC = 8 and BC = 6. We want to find $\sin(\angle BAC)$. In a right-angled triangle, the sine of an angle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. First, we need to find the length of the hypotenuse AB using the Pythagorean theorem: $AB^2 = AC^2 + BC^2$. $AB^2 = 8^2 + 6^2$ $AB^2 = 64 + 36$ $AB^2 = 100$ $AB = \sqrt{100} = 10$ units. Now, we can find $\sin(\angle BAC)$.

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45 [Challenging] • Target: Trigonometric ratios • Cross-Lesson: 9.03 The Binomial Theorem

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$ ✓

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

For any point on the unit circle, the coordinates are $(\cos(\theta), \sin(\theta))$. The equation of the unit circle is $x^2 + y^2 = 1$. Substituting the coordinates, we get $\cos^2(\theta) + \sin^2(\theta) = 1$. We are given that $\cos(\theta) = \frac{1}{2}$. So, $\left(\frac{1}{2}\right)^2 + \sin^2(\theta) = 1$. This simplifies to $\frac{1}{4} + \sin^2(\theta) = 1$. Subtracting $\frac{1}{4}$ from both sides gives $\sin^2(\theta) = 1 - \frac{1}{4} = \frac{3}{4}$. Taking the square root of both sides, we get.

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